CEE6501 — Lecture 8.1

Member Releases

Plane frames and beams with moment releases (hinges)

Learning Objectives

By the end of this lecture, you will be able to:

- Explain what a **member release** is and why it matters in DSM
- Classify members by **release type** (no hinge / hinge at b / hinge at e / hinges at both ends)
- Derive modified local stiffness relations by enforcing **zero end moment** at a hinge
- Write the resulting **modified local stiffness matrix** and **modified fixed-end force vector**
- Recognize and handle **hinged joints** and why they can cause singular stiffness matrices

Agenda

- **Part 1** — Concepts + notation
- **Part 2** — Derivations (MT = 1, 2, 3)
- **Part 3** — Hinged joints & Matrix Singularity
- **Part 4** — Method A, Fictitious Fixity
- **Part 5** — Method B, Fictitious Stiffness
- **Part 6** — Implementing Method A
- **Part 7** — In-Class Excercise
- **Part 8** — Implementing Method B

Part 1 — Concepts and Notation

What is a member release?

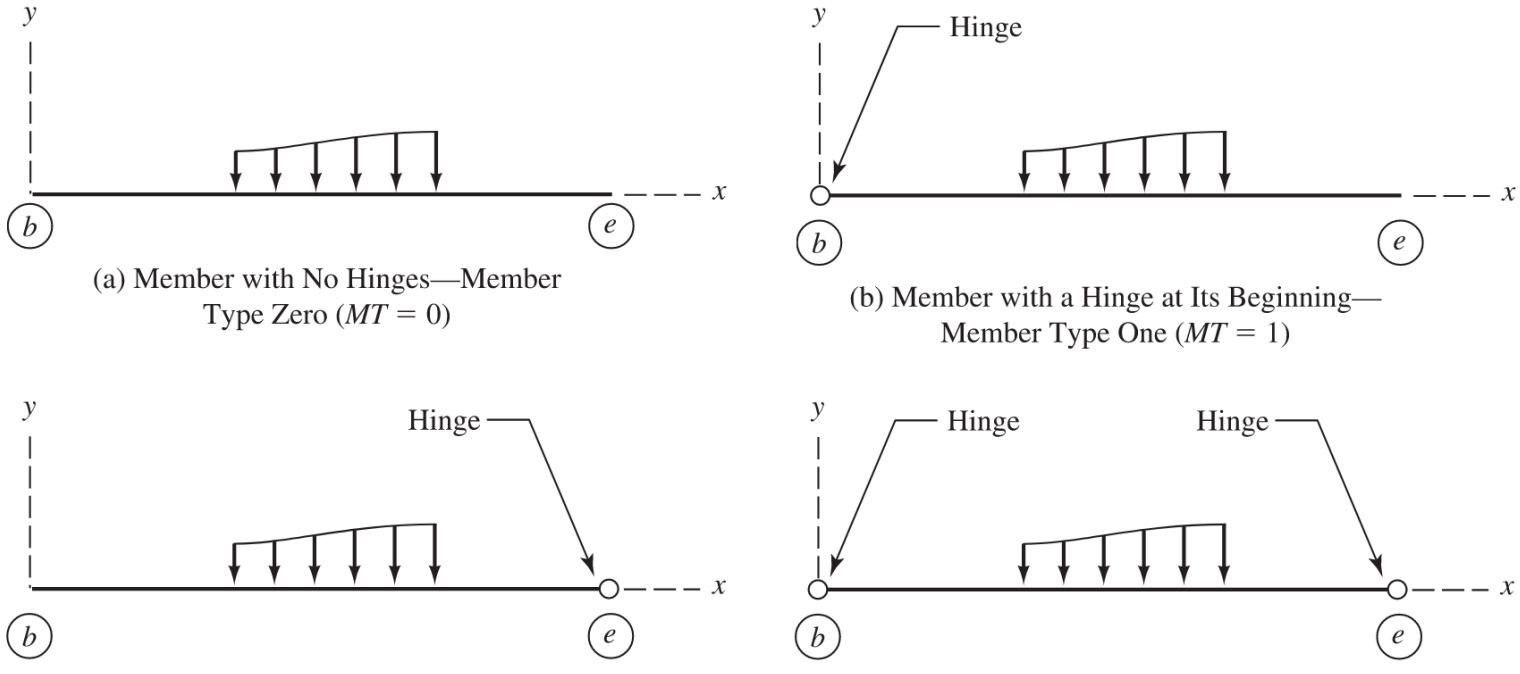
- A **member release** removes a force/moment transfer capability at an end of a member.
- Here we focus on **moment releases** (hinges).
- If an end is hinged, the **end moment must be zero** at that end.

Key constraint (local):

$$M_b = 0 \quad \text{or} \quad M_e = 0$$

Member Release Types (Classification)

- **MT = 0:** no hinges (rigid at both ends, normal frame element)
- **MT = 1:** hinge at **beginning** (end **b**)
- **MT = 2:** hinge at **end** (end **e**)
- **MT = 3:** hinges at **both ends**



(a) Member with No Hinges—Member Type Zero ($MT = 0$)

(b) Member with a Hinge at Its Beginning—Member Type One ($MT = 1$)

(c) Member with a Hinge at Its End—Member Type Two ($MT = 2$)

(d) Member with Hinges at Both Ends—Member Type Three ($MT = 3$)

Reminder: Frame Local DOFs and End Forces Notation

Local displacement vector (member coordinates):

$$\mathbf{u}^T = [\, u_1 \quad u_2 \quad u_3 \quad u_4 \quad u_5 \quad u_6 \,] = [\, u_b \quad v_b \quad \theta_b \quad u_e \quad v_e \quad \theta_e \,]$$

Local end-force vector:

$$\mathbf{Q}^T = [\, Q_1 \quad Q_2 \quad Q_3 \quad Q_4 \quad Q_5 \quad Q_6 \,] = [\, N_b \quad V_b \quad M_b \quad N_e \quad V_e \quad M_e \,]$$

Reminder: Frame Fixed-End Force Notation (FEFs)

We write local fixed-end forces using the convention:

- Q^F_{xb} = FEF at node **b** in local **x** (axial) → corresponds to **DOF 1**
- Q^F_{yb} = FEF at node **b** in local **y** (shear) → corresponds to **DOF 2**
- Q^F_{mb} = FEF at node **b** (moment) → corresponds to **DOF 3**
- Q^F_{xe} = FEF at node **e** in local **x** (axial) → corresponds to **DOF 4**
- Q^F_{ye} = FEF at node **e** in local **y** (shear) → corresponds to **DOF 5**
- Q^F_{me} = FEF at node **e** (moment) → corresponds to **DOF 6**

So the fixed-end force vector is

$$\mathbf{Q}^{F,T} = \begin{bmatrix} Q^F_{xb} & Q^F_{yb} & Q^F_{mb} & Q^F_{xe} & Q^F_{ye} & Q^F_{me} \end{bmatrix}^T$$

Part 2 — Derivations for Released Members

Starting point (MT = 0): standard local stiffness relation

$$\mathbf{Q} = \mathbf{k}\,\mathbf{u} + \mathbf{Q}^F$$

Member releases modify **only**:

- the local stiffness matrix $\mathbf{k}$, and
- the fixed-end force vector $\mathbf{Q}^F$

so the DSM workflow stays the same.

Base local stiffness relations (no hinges)

Recall from last lecture. For a plane frame member with no hinges, the local end-force components can be written (explicitly) as 6 independent equations:

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(12u_2 + 6Lu_3 - 12u_5 + 6Lu_6) + Q_{yb}^F \\
Q_3 &= \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6) + Q_{ye}^F \\
Q_6 &= \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F
\end{aligned}
$$

6 Independent Equations in Matrix Form

$$\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix} + \begin{Bmatrix} Q_1^F \\ Q_2^F \\ Q_3^F \\ Q_4^F \\ Q_5^F \\ Q_6^F \end{Bmatrix}$$

MT = 0: Matrix Form

$$\mathbf{k}_{\text{frame}} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix}$$

MT = 1 (hinge at b): enforce $M_b = 0$

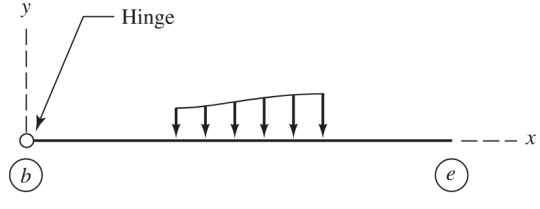
Hinge at the beginning means

$$M_b = Q_3 = 0$$

Start from

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Set $Q_3 = 0$ and solve for u_3 (released rotation at b).



MT = 1: solve for the released rotation u_3

Set $Q_3 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_3:

$$4L^2u_3 = -6L(u_2 - u_5) - 2L^2u_6 - \frac{L^3}{EI}Q_{mb}^F$$

$$u_3 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_6 - \frac{L}{4EI}Q_{mb}^F$$

Key point: *u_3 is no longer an independent DOF in the member relations. The row/column corresponding to this DOF can be expressed in terms of a linear combination of other rows/columns in the matrix.*

MT = 1: substitute u_3 back into Q_2, Q_5, Q_6

Resulting modified equations (MT = 1):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 - 3u_5 + 3Lu_6) + \left(Q_{yb}^F - \frac{3}{2L}Q_{mb}^F\right) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 + 3u_5 - 3Lu_6) + \left(Q_{ye}^F + \frac{3}{2L}Q_{mb}^F\right) \\
Q_6 &= \frac{EI}{L^3}\left(3Lu_2 - 3Lu_5 + 3L^2u_6\right) + \left(Q_{me}^F - \frac{1}{2}Q_{mb}^F\right)
\end{aligned}
$$

MT = 1: stiffness matrix

row/column 3 are 0's:

$$\mathbf{k}_{(MT=1)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & \boxed{0} & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 3 & \boxed{0} & 0 & -3 & 3L \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -\frac{AL^2}{I} & 0 & \boxed{0} & \frac{AL^2}{I} & 0 & 0 \\ 0 & -3 & \boxed{0} & 0 & 3 & -3L \\ 0 & 3L & \boxed{0} & 0 & -3L & 3L^2 \end{bmatrix}$$

MT = 1: fixed end force vector

row/column 3 are 0's:

$$\mathbf{Q}^F_{(MT=1)} = \begin{bmatrix} F^F_{xb} \\ F^F_{yb} - \frac{3}{2L}F^F_{mb} \\ \boxed{0} \\ F^F_{xe} \\ F^F_{ye} + \frac{3}{2L}F^F_{mb} \\ F^F_{me} - \frac{1}{2}F^F_{mb} \end{bmatrix}$$

MT = 2 (hinge at e): enforce $M_e = 0$

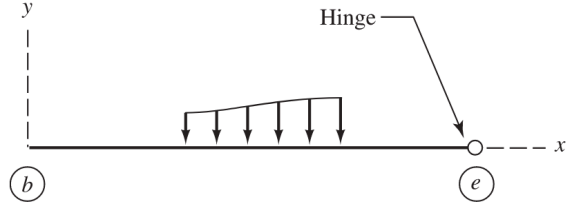
Hinge at the end means

$$M_e = Q_6 = 0$$

Start from

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Set $Q_6 = 0$ and solve for u_6 (released rotation at e).



MT = 2: solve for the released rotation u_6

Set $Q_6 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_6:

$$4L^2u_6 = -6L(u_2 - u_5) - 2L^2u_3 - \frac{L^3}{EI}Q_{me}^F$$

$$u_6 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_3 - \frac{L}{4EI}Q_{me}^F$$

MT = 2: substitute u_6 back into Q_2, Q_3, Q_5

Resulting modified equations (MT = 2):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 + 3Lu_3 - 3u_5) + \left(Q_{yb}^F - \frac{3}{2L}Q_{me}^F\right) \\
Q_3 &= \frac{EI}{L^3}\left(3Lu_2 + 3L^2u_3 - 3Lu_5\right) + \left(Q_{mb}^F - \frac{1}{2}Q_{me}^F\right) \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 - 3Lu_3 + 3u_5) + \left(Q_{ye}^F + \frac{3}{2L}Q_{me}^F\right) \\
Q_6 &= 0
\end{aligned}
$$

MT = 2: stiffness matrix

Row and column 6 are zero:

$$\mathbf{k}_{(MT=2)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & 3 & 3L & 0 & -3 & \boxed{0} \\ 0 & 3L & 3L^2 & 0 & -3L & \boxed{0} \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & -3 & -3L & 0 & 3 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

MT = 2: fixed-end force vector

Row/column 6 are zero:

$$\mathbf{Q}_{f,(MT=2)} = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \dfrac{3}{2L} Q_{me}^F \\ Q_{mb}^F - \dfrac{1}{2} Q_{me}^F \\ Q_{xe}^F \\ Q_{ye}^F + \dfrac{3}{2L} Q_{me}^F \\ \boxed{0} \end{bmatrix}$$

MT = 3 (hinges at both ends): enforce $M_b = 0$ and $M_e = 0$

Hinges at both ends mean

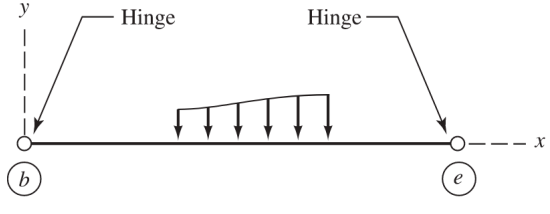
$$Q_3 = 0, \qquad Q_6 = 0$$

We now have **two constraints** that couple u_3 and u_6.

Start from the base moment equations:

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$



MT = 3: solve the coupled system for u_3 and u_6

Set $Q_3 = 0$ and $Q_6 = 0$, multiply both by $\frac{L^3}{EI}$:

$$0 = 6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6 + \frac{L^3}{EI}Q^F_{mb}$$

$$0 = 6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6 + \frac{L^3}{EI}Q^F_{me}$$

Solving gives (one convenient form):

$$u_3 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{mb} - Q^F_{me})$$

$$u_6 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{me} - Q^F_{mb})$$

MT = 3: substitute u_3 and u_6 back Q_2, Q_5

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= Q_{yb}^F - \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= Q_{ye}^F + \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_6 &= 0
\end{aligned}
$$

Interpretation

- Notice that Q_2 **and** Q_5 **no longer contain any displacement terms**.
- They are **purely functions of the fixed-end forces**.
- The bending stiffness contributions have completely disappeared.
- This means the element **cannot develop bending resistance**.
- The transverse shear forces are determined solely by equilibrium of the fixed-end moments.

MT = 3: stiffness matrix

Not only are row/column 3 and 6 are zero, but also row/column 2 and 5:

$$\mathbf{k}_{(MT=3)} = \frac{EA}{L}\begin{bmatrix} 1 & 0 & \boxed{0} & -1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -1 & 0 & \boxed{0} & 1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

What does this stiffness matrix remind you of?

- The element now behaves as a **pure axial (truss-type) member**.

MT = 3: fixed-end force vector

Row/column 3 and 6 are zero:

$$\mathbf{Q}^F_{(MT=3)} = \begin{bmatrix} Q^F_{xb} \\ Q^F_{yb} - \frac{1}{L}(Q^F_{mb} + Q^F_{me}) \\ \boxed{0} \\ Q^F_{xe} \\ Q^F_{ye} + \frac{1}{L}(F^F_{mb} + Q^F_{me}) \\ \boxed{0} \end{bmatrix}$$

Implementation note: keep 6×6, but released DOFs have zero rows/cols

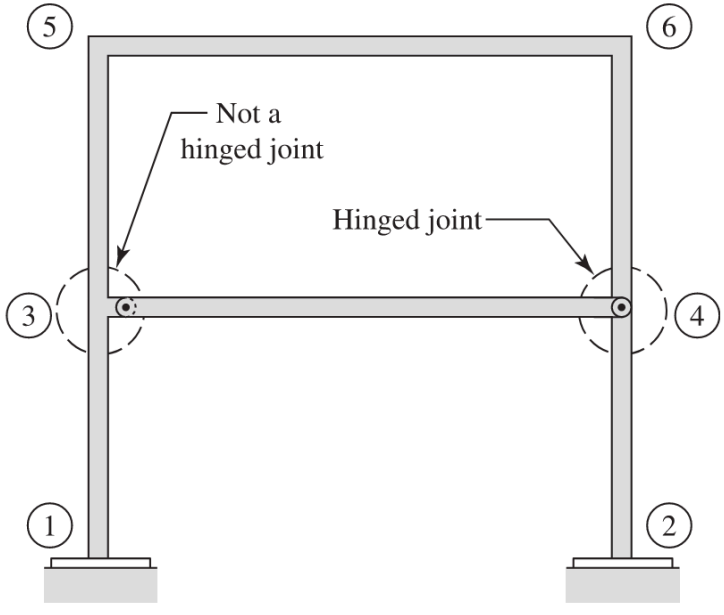
- Even though releases reduce the **independent** member coordinates, we often keep a **6×6** member form.
- Released moment rows/cols (and their corresponding fixed-end terms) become **zero** in the stiffness matrix and FEF vector.
- This keeps the DSM assembly code clean: same DOF mapping size for every member.

Then transform each element to global coordinates as usual:

$$\mathbf{K} = \mathbf{T}^T \mathbf{k} \mathbf{T}, \qquad \mathbf{F}^F = \mathbf{T}^T \mathbf{Q}^F$$

Part 3 — Hinged Joints and Matrix Singularity

Hinged joints in beams and plane frames



A **hinged joint** is a joint where **ALL** members meeting at that joint are connected by **moment releases**.

Physical consequence:

- The joint **cannot transmit bending moment**
- The joint is **free to rotate**
- Axial and shear forces may still be transmitted

What does this mean for the DOFs?

At a hinged joint:

- The rotational DOF θ_j is not constrained (it is free).
- But no connected member provides **rotational stiffness**.
- The joint rotation does not generate bending moment.

In other words: The structure has a rotation that costs **zero strain energy**.

DSM partitioned equilibrium

Recall the partitioned system:

$$\begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{F}_f \\ \mathbf{F}_r \end{bmatrix}$$

We solve for the unknown displacements from

$$\mathbf{u}_f = \mathbf{K}_{ff}^{-1}(\mathbf{F}_f - \mathbf{K}_{fr}\mathbf{u}_r)$$

This requires that $\mathbf{K}_{ff}$ be **invertible**.

What happens if θ_j is left in the free set?

The rotational DOF θ_j is not restrained, so it is placed in the free displacement vector $\mathbf{u}_f$.

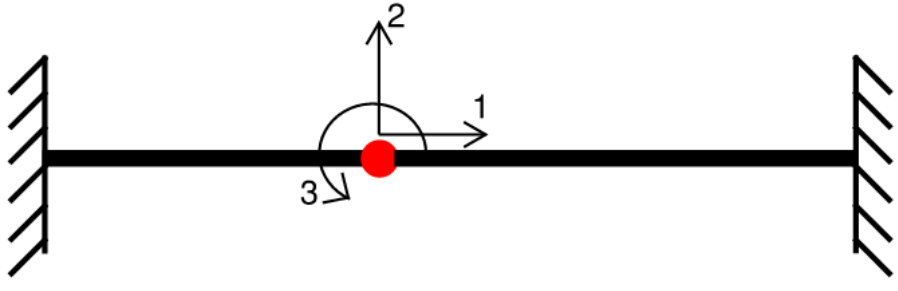
As a free DOF, it has a corresponding row and column in the stiffness matrix $\mathbf{K}_{ff}$.

However, at a fully hinged joint, no connected element provides rotational stiffness. From our earlier derivations, the entire row and column associated with θ_j are zero.

Therefore:

- $\mathbf{K}_{ff}$ contains a **zero row and column**.
- $\mathbf{K}_{ff}$ becomes **singular** (non-invertible).
- The system of equations does not admit a unique displacement solution.

Simple Example with Joint DOF Included in $\mathbf{K}_{ff}$



$$\mathbf{K}_{ff} = \begin{bmatrix} k_{11} & k_{12} & 0 \\ k_{21} & k_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Important clarification

This situation does **not** indicate a physical instability.

The structure may be completely stable in translation and capable of carrying load.

The singularity arises because a rotational DOF was left in the free set **without any associated stiffness**.

In other words:

- The joint can rotate freely.
- That rotation does not generate internal moment.
- The stiffness matrix has no resistance in that direction.

The issue is therefore **mathematical**, not structural.

To obtain a well-posed system of equations, this DOF must be treated properly in the DSM formulation.

Two Modeling Approaches

Two practical ways to avoid singularity:

Method A — Artificially restrain (fix) the joint rotation

- Put the hinged-joint rotation into the **restrained set** $\mathbf{u}_r$ (not in $\mathbf{u}_f$)
- The computed reaction moment at that DOF ends up being **zero** anyway

Method B — Artifically stiffen the joint DOF

- Model the joint as rigidly connected to **ONLY ONE** of the members meeting there
- Keep the joint DOFs (u_x, u_y, θ) free as usual
- Other members at the joint are connected by hinges (moment releases) so the net joint moment remains zero

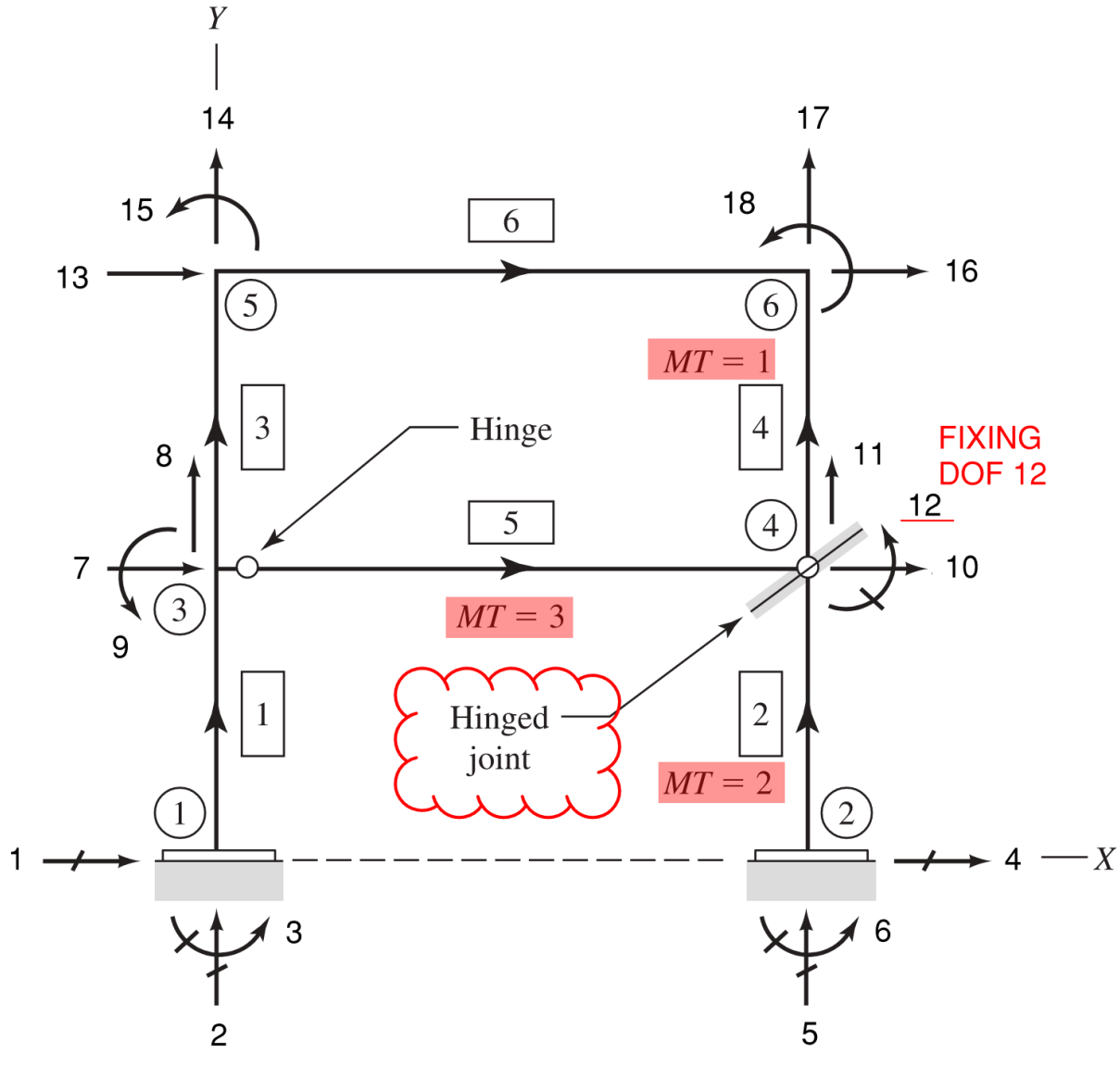
Part 4 - Method A, Fictitious Fixity

The Main Idea

The most straightforward remedy for the singularity is:

> *Eliminate the rotational DOF of hinged joints by modeling such joints as restrained (fixed) against rotation.*

This is imposing **fictitious fixity** that does not change global structural behavior.


Y
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17
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MT = 1
3
Hinge
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FIXING
DOF 12
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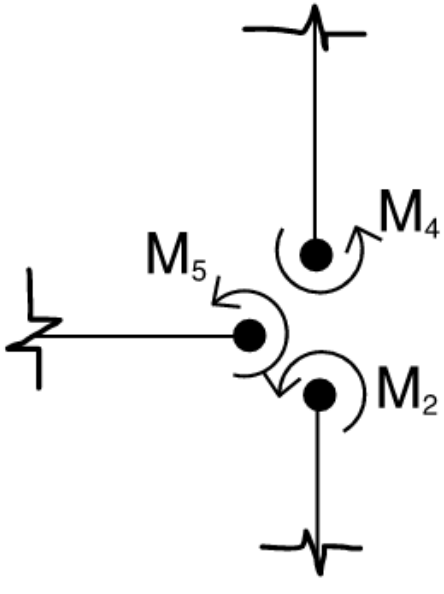
7
10
3
MT = 3
9
1
Hinged
joint
2
MT = 2
1
2
1
4
X
3
6
2
5

Internal Moment at a Fully Hinged Joint

At a fully hinged joint:

- All connected members have **moment releases** at the node.
- The joint **cannot transmit bending moment**.
- Therefore, each connected element, i, contributes

$$M_i = 0 \quad \text{at the joint.}$$



Artificially Fixing the Rotation

Suppose we introduce **fictitious fixity** by restraining the rotation at the joint DOF associated with joint j:

$$\theta_j = 0$$

Instead of leaving θ_j in the free displacement vector, we move it into the restrained set.

After solving for the free DOFs, the reaction vector, F_r, is computed containing the load value for all restrained DOFs.

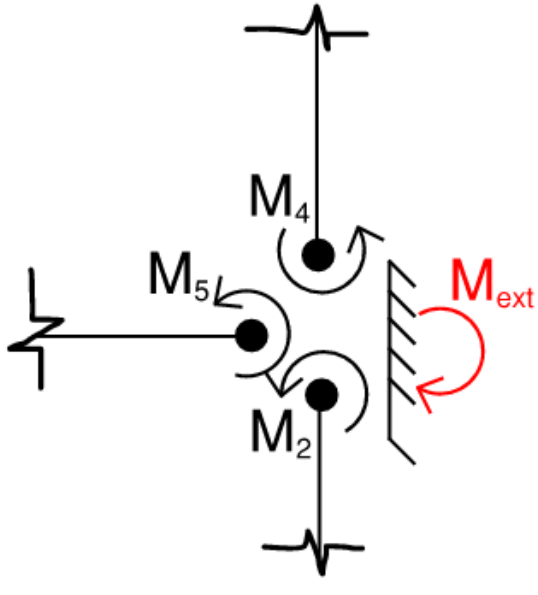
For the fictitiously fixed rotation at node j, let us call the reported reaction moment: M_{ext}

Joint equilibrium must always satisfy

$$\sum_{i=1}^{n_{ele}} M_i + M_{\text{ext}} = 0$$

where:

- M_i = internal end moment from element i
- M_{ext} = applied or reaction moment at the joint



What Happens to the "Reaction" at That DOF?

At a **hinged joint**, all connected members have **moment releases**, so none of them can transmit bending moment to the joint.

Therefore the joint moment equilibrium becomes:

$$\sum_{i=1}^{n_{ele}} M_i = 0$$

And since no element can deliver moment to the joint:

$$M_{\text{ext}} \approx 0$$

Thus, the reported "reaction" at the fictitiously fixed joint, j, is a numerical artifact must evaluates to zero (up to roundoff).

Why the Structural Response is Unchanged

Imposing $\theta_j = 0$ does not change the moments developed in the joint, because:

- No connected element provides rotational stiffness.
- The global stiffness matrix contains no coupling terms involving θ_j.
- The equilibrium equations do not depend on that rotational DOF.

Because moment is related to rotation through stiffness, and the rotational stiffness is zero, restraining the rotation cannot generate a moment.

Thus, even if the rotation is **artificially fixed**, the joint develops **no reaction moment**, and the structural response remains unchanged.

You can think of θ_j as a restrained DOF that simply develops **no reaction load**.

Implementing with DSM Partitioning

Instead of leaving θ_j in the free displacement vector $\mathbf{u}_f$,

we classify it as a **restrained DOF**.

So:

- It is placed in $\mathbf{u}_r$.
- It is removed from $\mathbf{K}_{ff}$.
- It is no longer part of the system we invert:

$$\mathbf{K}_{ff}^{-1}$$

The zero-stiffness row and column disappear from the matrix being solved.

Method A Summary

Method A is therefore a **numerical bookkeeping device**:

- It removes a zero-stiffness DOF from the unknown set,
- Eliminates the singularity in $\mathbf{K}_{ff}$,
- Preserves the correct internal force distribution.

The physics is unchanged but the algebra becomes well-posed.

Method A — Step-by-Step Procedure

1. **Identify the hinged joint**, j

    - All connected member ends at that node are moment-released.
    - The joint cannot transmit bending moment.

2. **Classify the rotational DOF as restrained**

    - Place θ_j in the restrained displacement vector $\mathbf{u}_r$.
    - Impose the artificial condition

$$\theta_j = 0$$

This removes the hinge rotation from the unknown set, u_f.

3. **Assemble the global stiffness matrix normally**

    - Because all connected elements are released, the assembled stiffness terms associated with θ_j are zero.
    - No rotational stiffness is introduced.

4. **Solve for the free DOFs**

    - θ_j is not included in $\mathbf{u}_f$.
    - The matrix $\mathbf{K}_{ff}$ no longer contains a zero row/column.
    - The system becomes invertible.

5. **Compute reactions**

    Reactions are obtained from

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F$$

    The reaction moment at the fixed rotational DOF at joint j evaluates to:

$$M_j \approx 0$$

    confirming hinge behavior.

Part 5 - Method B, Fictitious Stiffness

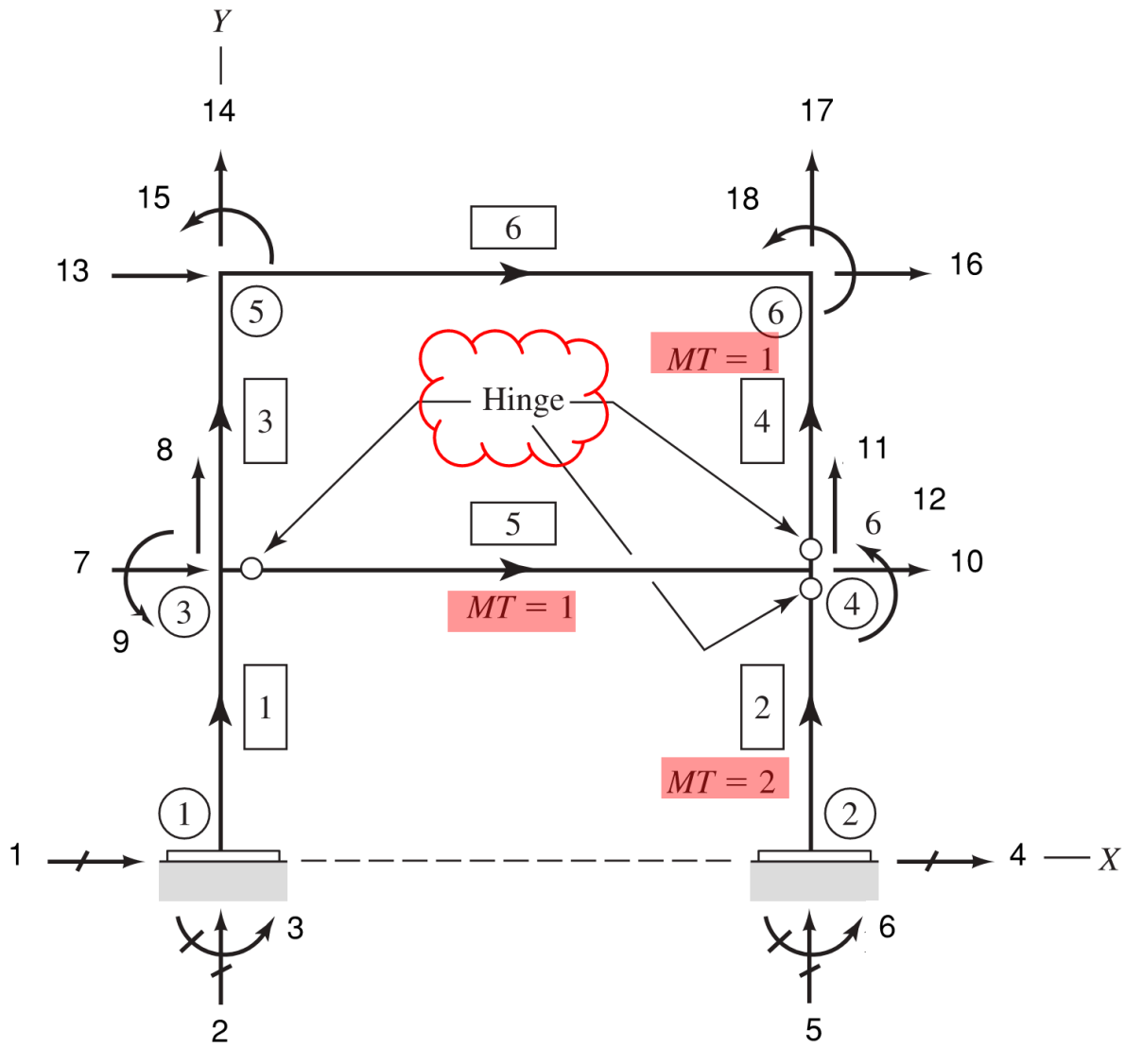
The Main Idea

Instead of removing the joint rotation from the free system (Method A), we:

> *Model **one (and only one)** of the members meeting at that joint as having rotational rigidity.*

All other members framing into the joint remain **hinged (moment-released)**.

This is imposing **fictitious rigidity** that does not change global structural behavior.


Y
14
17
15
18
6
13
16
5
6
MT = 1
Hinge
3
4
8
11
12
5
6
7
10
3
4
9
MT = 1
1
2
MT = 2
1
2
1
4
X
3
6
2
5

Why This Removes the Singularity

By making **one member rigidly connected**, that member contributes:

$$\frac{4EI}{L}$$

type rotational stiffness at the joint.

Therefore:

- The joint rotation DOF now has stiffness.
- $\mathbf{K}_{ff}$ is no longer singular.
- The system becomes solvable without fictitious restraints.

Why This Does NOT Change the Structural Behavior

From joint equilibrium:

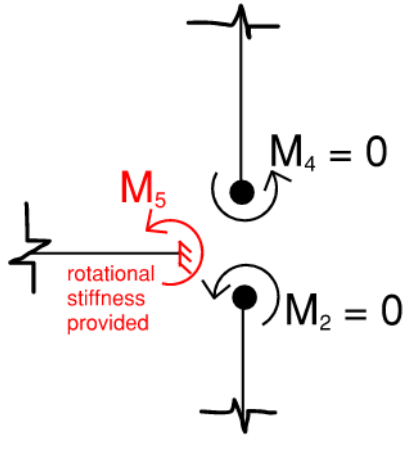
$$\sum_{i=1}^{n_{ele}} M_i = 0$$

Even though one member, j, is fictitiously modeled as rigid at the joint:

- All other members connected there are hinged.
- Hinged ends cannot transmit moment.

All hinged members satisfy $M_i = 0$, so the joint moment equilibrium can be written as:

$$\sum_{i=1}^{n_c} M_i = \left(\sum_{\substack{i=1 \\ i \neq j}}^{n_c} M_i \right) + M_j = 0 + M_j = 0$$



Thus, the rigidly connected member must also have zero end moment, $M_j \approx 0$, in the final solution in order to satisfy joint equilibrium.

Key Mechanical Insight

Although we introduced rotational stiffness at the joint:

- The structure has no moment demand at that location.
- Static equilibrium forces the joint moment to zero.
- The internal force distribution remains unchanged.

We have:

- Stabilized the algebra (no singular matrix),
- Preserved correct statics,
- Avoided artificial restraints.

What Method B Really Is

Method B is a **modeling device** that:

- Supplies stiffness to a mechanism DOF,
- Keeps the joint rotation as a free unknown,
- Relies on equilibrium to enforce zero moment at the joint (i.e., correct hinge behavior).

The structural response remains the same. The numerical system becomes well-posed.

Method B — Step-by-Step Procedure

1. **Identify the fully hinged joint**
    - Confirm that all members meeting at the joint are moment-released.
    - Recognize that this creates a rotational DOF with zero stiffness.
2. **Select one "reference" member at the joint**
    - Choose **one (and only one)** connected member.
    - Model that member as **rigidly connected** at the joint (no moment release at that end).
    - The rotation DOF is now supported by this member's rotational stiffness.

   This supplies stiffness to the otherwise unsupported rotation.
3. **Keep all other members hinged at that joint**
    - Use the appropriate released-end formulation (MT = 1 or MT = 2).
    - Their end moments at the joint are explicitly zero.

   These members do **not** contribute rotational stiffness at the joint.

4. **Treat the joint as a normal rigid node in global bookkeeping**

    - The rotational DOF θ_j remains in the **free displacement vector**.
    - Because one member now contributes stiffness, the corresponding row/column in $\mathbf{K}_{ff}$ is no longer zero.
    - The singularity disappears.

5. **Assemble and solve the system**

    - Assemble the global stiffness matrix in the usual way.
    - Solve for the free DOFs.
    - Compute member end forces.

Although one member was modeled as rigid at the joint, equilibrium at that joint still requires:

$$\sum M = 0$$

Since the other members are hinged and no external moment is applied, the rigid member's end moment evaluates to:

$$M_j \approx 0$$

Method A vs Method B — Clear Takeaway

Method A — Fictitious Fixity

- Reclassify the joint rotation DOF as **restrained**.

Effect:

- Eliminates the zero-stiffness DOF from $\mathbf{K}_{ff}$.
- The computed fictituous reaction moment evaluates to ≈ 0

Method B — Fictitious Stiffness

- Keep the joint rotation DOF **free**.
- Make **one (and only one)** connected member rigid at the joint.
- All other connected members remain hinged.

Effect:

- The rigid member supplies nonzero rotational stiffness.
- From joint equilibrium, $\sum M = 0$, the fictituously rigid member's end moment evaluates to ≈ 0.

What Is Different?

- **Method A:** enforces $\theta_j = 0$
- **Method B:** allows $\theta_j \neq 0$ (determined numerically from solving u_f)

Therefore, local element rotations can be slightly different across methods, but this does not affect the internal force distribution.

Key Takeaway

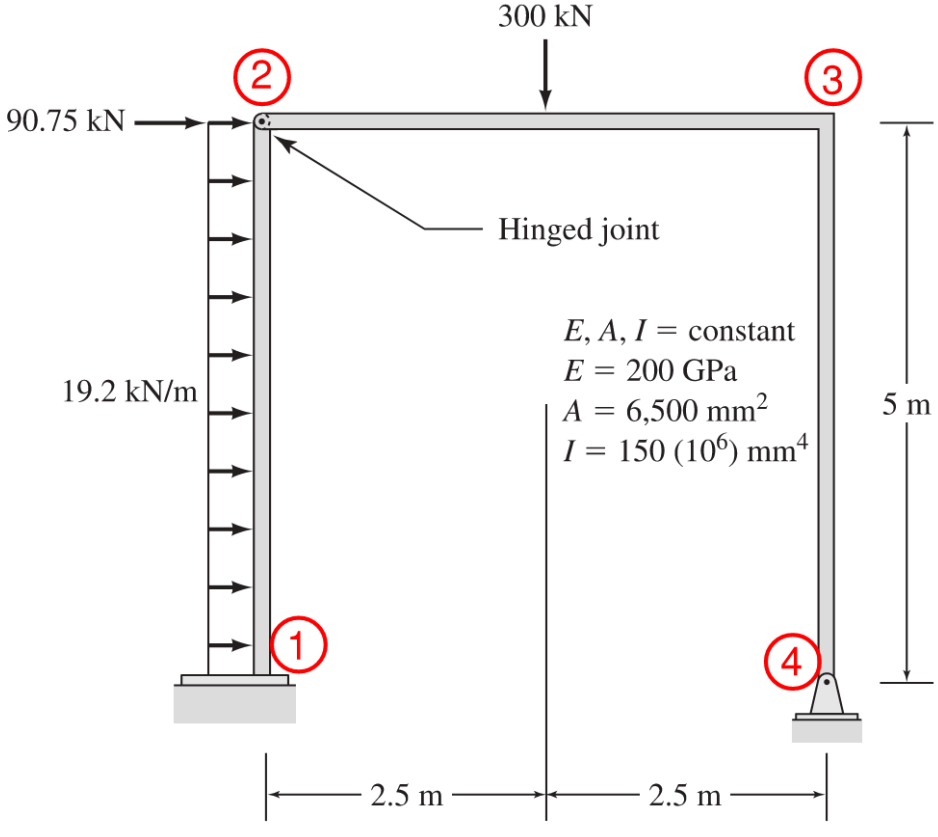
Both methods represent the **same physical structure** and satisfy the same equilibrium and compatibility conditions.

- Internal force distribution → **the same**
- Physically meaningful deformations → **the same**
- Reported joint rotation at the hinges → **may differ**

The difference is numerical, not structural.

Part 6 — Implementing Method A

Example Problem (7.1 Kassimali)



Problem data

Assume member properties are constant for all members:

- $E = 200 \times 10^6 \text{ kN/m}^2 \ (= 200 \text{ GPa})$
- $A = 0.0065 \text{ m}^2 \ (= 6500 \text{ mm}^2)$
- $I = 150 \times 10^{-6} \text{ m}^4 (= 150 \times 10^6 \text{ mm}^4)$
- $L = 5 \text{ m}$

Local DOF order for each 2D frame member:

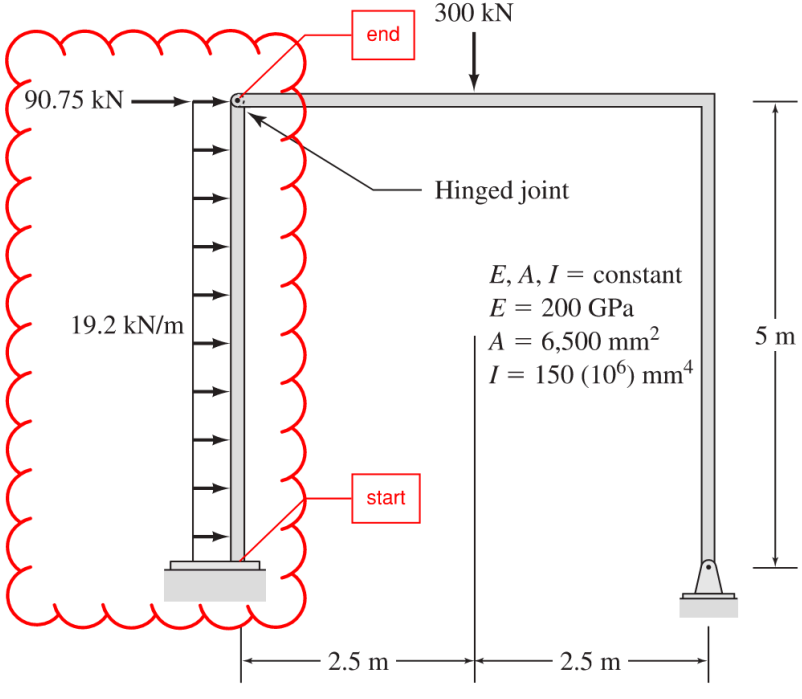
$$[u_{xb},\ u_{yb},\ \theta_b,\ u_{xe},\ u_{xe},\ \theta_e]$$

Fixed-end force notation:

$$\begin{aligned} \text{Node } b: &\quad Q^F_{xb},\ Q^F_{yb},\ Q^F_{mb} \\ \text{Node } e: &\quad Q^F_{xe},\ Q^F_{ye},\ Q^F_{me} \end{aligned}$$

Member 1 (MT = 2)

Member 1 has a **hinge at end** *e*, so it is **MT = 2**.



All stiffness terms are expressed in **kN** and **m**.

Local stiffness matrix (MT = 2):

$$\mathbf{k}_1 = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 3{,}600 & 0 & -720 & 0 \\ 0 & 3{,}600 & 18{,}000 & 0 & -3{,}600 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & -3{,}600 & 0 & 720 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Local fixed-end forces for uniform transverse load $w = 19.2\ \mathrm{kN/m}$:

First, fully-fixed (rigid) FEF component.

No inclined force or member, axial = 0:

$$Q_{xb}^F = Q_{xe}^F = 0$$

Shear and moment:

$$Q_{yb}^F = Q_{ye}^F = \frac{wL}{2} = \frac{19.2(5)}{2} = 48\ \mathrm{kN}$$

$$Q_{mb}^F = -Q_{me}^F = \frac{wL^2}{12} = \frac{19.2(5)^2}{12} = 40\ \mathrm{kN\text{-}m}$$

Then modify for MT = 2 (hinge at end e):

$$\mathbf{Q}_1^F = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \frac{3}{2L} F_{me}^F \\ Q_{mb}^F - \frac{1}{2} F_{me}^F \\ Q_{xe}^F \\ Q_{ye}^F + \frac{3}{2L} F_{me}^F \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 60 \\ 60 \\ 0 \\ 36 \\ 0 \end{bmatrix}$$

Transformation matrix: member 1 is vertical ($\theta = 90^\circ$, $c = 0$, $s = 1$)

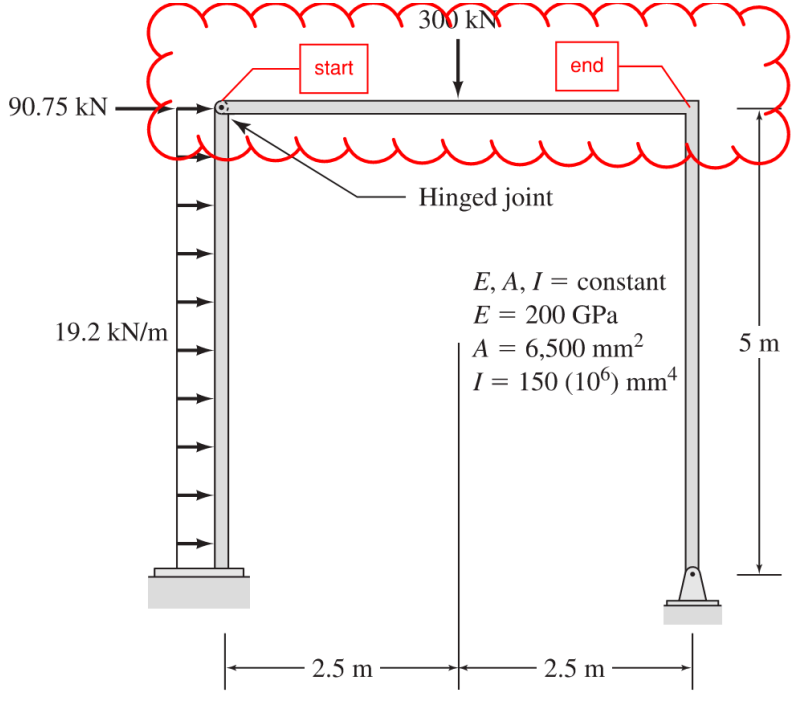
$$\mathbf{T}_1 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [1, 2, 3, 4, 5, 6]_{\text{global}}$$

Member 2 (MT = 1)

Member 2 has a **hinge at end** b, so it is **MT = 1**.



All stiffness terms are expressed in **kN** and **m**.

Local stiffness matrix (MT = 1):

$$\mathbf{k}_2 = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 0 & 0 & -720 & 3{,}600 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & 0 & 0 & 720 & -3{,}600 \\ 0 & 3{,}600 & 0 & 0 & -3{,}600 & 18{,}000 \end{bmatrix}$$

Local fixed-end forces (point load case shown in the example):

Rigid-member (fully fixed) FEF components used in the example:

$$Q_{xb}^F = Q_{xe}^F = 0$$

$$Q_{yb}^F = Q_{ye}^F = 150\text{kN}$$

$$Q_{mb}^F = -Q_{me}^F = 187.5\text{kN-m}$$

Modify for MT = 1 (hinge at end b):

$$\mathbf{Q}_2^F = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \frac{3}{2L} Q_{mb}^F \\ 0 \\ Q_{xe}^F \\ Q_{ye}^F + \frac{3}{2L} Q_{mb}^F \\ Q_{me}^F - \frac{1}{2} Q_{mb}^F \end{bmatrix} = \begin{bmatrix} 0 \\ 93.75 \\ 0 \\ 0 \\ 206.25 \\ -281.25 \end{bmatrix}$$

Transformation matrix: member 2 is horizontal (aligned with global), so

$$\mathbf{T}_2 = \mathbf{I}_{6\times 6}$$

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [2, 3, 4, 5, 6, 7]_{\text{global}}$$

Member 3 (MT = 1)

At the start of member 3 there is a **pin support**.

We have two equivalent modeling choices:

Option 1 — Standard Frame Element (MT = 0)

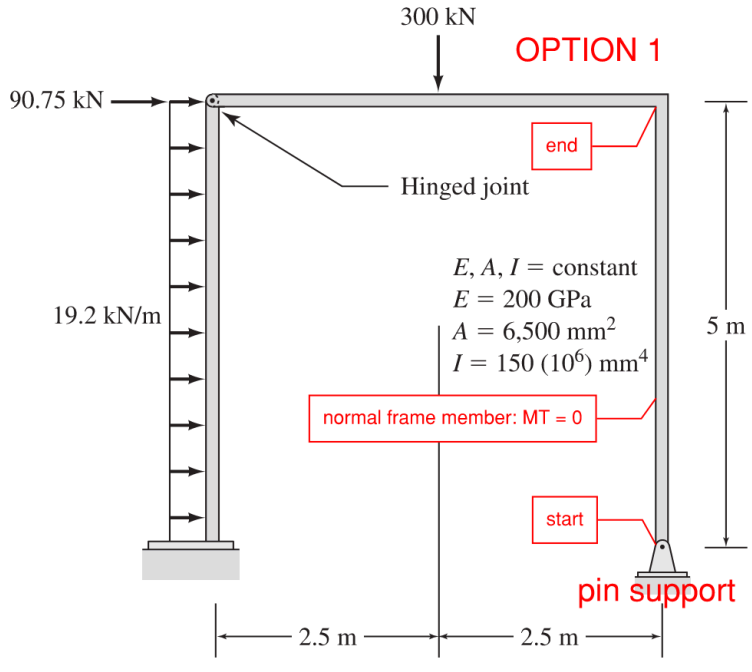
- Use the full 6×6 local stiffness matrix (no releases)
- Impose $u_x = 0$ and $u_y = 0$ in the restrained DOF set
- Leave the rotational DOF θ free

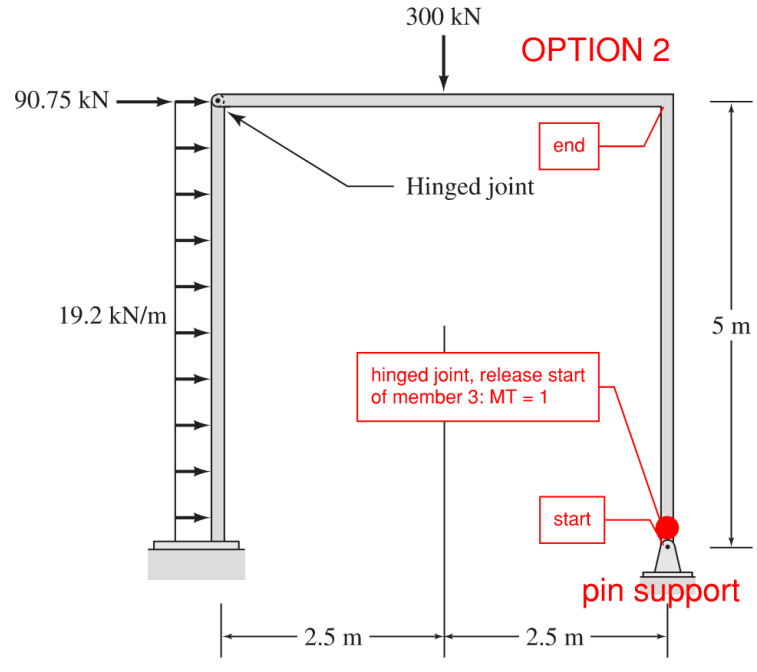
In this approach, this location remains numerically stable without any additional treatment.

Option 2 — Member Release at Start (MT = 1)

- The pin support removes moment transfer at that end of the element
- This is the same effect as if there was a **hinge** at the start of the element
- The element can therefore be classified as **MT = 1**

Because a rotational release eliminates bending stiffness at that end, this creates another situation where fictitous fixity must be applied


300 kN
OPTION 1
90.75 kN
end
Hinged joint
E, A, I = constant
E = 200 GPa
A = 6,500 mm^2
I = 150 (10^6) mm^4
19.2 kN/m
5 m
normal frame member: MT = 0
start
pin support
2.5 m
2.5 m


300 kN
OPTION 2
90.75 kN
end
Hinged joint
19.2 kN/m
5 m
hinged joint, release start
of member 3: MT = 1
start
pin support
2.5 m
2.5 m

In this example, Option 2 is convenient because:

- Geometrically, member 3 matches member 2.
- Same release condition (hinge at b).
- So we can reuse:

$$\mathbf{k}_3 = \mathbf{k}_2$$

- No member loading:

$$\mathbf{Q}_3^F = \mathbf{0}$$

- Vertical member:

$$\theta = 90^\circ \quad \mathbf{T}_3 = \mathbf{T}_1$$

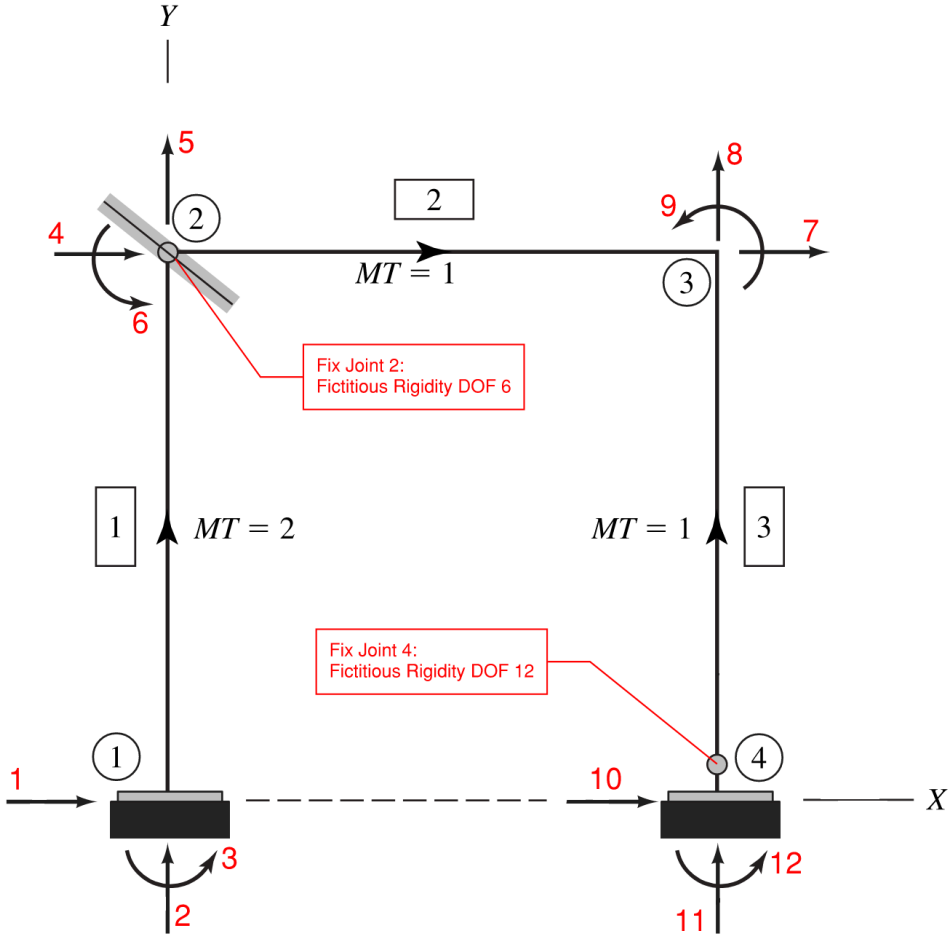
- Mapping

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [10, 11, 12, 7, 8, 9]_{\text{global}}$$

Note: Both approaches are correct — using MT = 1 simply keeps the setup cleaner.

You will prove this later using a standard 6x6 element (in class excercise)

Fictitious Fixity at Nodes 2 and 4



In [49]:

```python
# -------------------------
# Import Libraries and Helper Functions
# -------------------------
# This slide imports the required Python libraries and adds the
# course helper-code directory to the Python path.

import sys
import os
import numpy as np
import pandas as pd

# Get current notebook directory
current_dir = os.getcwd()

# Go up two levels, then into Code/L8
helpers_path = os.path.abspath(
    os.path.join(current_dir, "..", "..", "Code", "L8")
)

sys.path.append(helpers_path)
```

In [50]:
```
from helper_funcs_dsm import (
    assemble_global_stiffness_and_fef,
    partition_system,
    assemble_global_displacements,
    assemble_global_forces
)
from helper_funcs_output import (
    print_element,
    print_dsm_results,
    print_matrix_scaled
)

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6          # kPa
A = 0.0065         # m^2
I = 150e-6         # m^4
L = 5.0            # m
```

In [51]:

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [52]:

```
# -------------------------
# Member 2 (MT=1)
# -------------------------
k2 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,     0.,      0.,  -720.,  3600.],
    [    0.,     0.,     0.,      0.,     0.,     0.],
    [-260000.,    0.,     0.,  260000.,     0.,     0.],
    [    0.,  -720.,     0.,      0.,   720., -3600.],
    [    0.,  3600.,     0.,      0., -3600., 18000.]
], dtype=float)

# Local fixed-end vector for member 2 (example result)
Qf2 = np.array([0., 93.75, 0., 0., 206.25, -281.25], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [53]:

```
# -------------------------
# Member 3 (MT=1)
# -------------------------
k3 = k2.copy()
Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [54]:

```
# -------------------------
# Initialize Global System
# -------------------------
# This slide sets up the global variables needed for assembly of the
# Direct Stiffness Method system.

k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))

# Initialize Applied Load
F_global = np.zeros(ndof, dtype=float)

# Initialize Global Displacement
u_global = np.zeros(ndof, dtype=float)
```

In [55]:
```
# -------------------------
# User-Specified Boundary Conditions
# -------------------------
# This slide defines the external loads and support conditions.

# Applied Load
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([6, 12], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)

print(dof_restrained_1based)
```

```
[ 1  2  3  6 10 11 12]
```

In [56]:

```
# -------------------------
# Assemble Global Stiffness System
# -------------------------
# This step assembles the global stiffness matrix and the global
# fixed-end force vector from all elements.

K_global, F_fef_global = assemble_global_stiffness_and_fef(
    ndof,
    k_list,
    T_list,
    Qf_list,
    map_list
)
```

In [57]:

```
# -------------------------
# Display Global Stiffness Matrix
# -------------------------
# Print the assembled global stiffness matrix in a readable format.
# The matrix is scaled for clearer presentation in the slides.

print_matrix_scaled(K_global, scale=1000, decimals=1, col_width=4)
```

```
K = 1e+03 ×
01 |  0.7  0.0 -3.6 -0.7  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0
02 |  0.0 260.0  0.0  0.0 -260.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0
03 | -3.6  0.0 18.0  3.6  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0
04 | -0.7  0.0  3.6 260.7  0.0  0.0 -260.0  0.0  0.0  0.0  0.0  0.0
05 |  0.0 -260.0  0.0  0.0 260.7  0.0  0.0 -0.7  3.6  0.0  0.0  0.0
06 |  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0
07 |  0.0  0.0  0.0 -260.0  0.0  0.0 260.7  0.0  3.6 -0.7  0.0  0.0
08 |  0.0  0.0  0.0  0.0 -0.7  0.0  0.0 260.7 -3.6  0.0 -260.0  0.0
09 |  0.0  0.0  0.0  0.0  3.6  0.0  3.6 -3.6 36.0 -3.6  0.0  0.0
10 |  0.0  0.0  0.0  0.0  0.0  0.0 -0.7  0.0 -3.6  0.7  0.0  0.0
11 |  0.0  0.0  0.0  0.0  0.0  0.0  0.0 -260.0  0.0  0.0 260.0  0.0
12 |  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0
```

In [58]:

```
# -------------------------
# Partition Global System
# -------------------------
# The global stiffness equation is partitioned into free and restrained
# degrees of freedom based on the support conditions.

(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    u_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    u_global,
    F_fef_global,
    dof_restrained_1based
)
```

$$u_f = K_{ff}^{-1}(F_f - F_f^F - K_{fr}u_r)$$

$$F_r = K_{rf}u_f + K_{rr}u_r + F_r^F$$

In [59]:

```
# -------------------------
# Solve Partitioned System
# -------------------------
# Solve for the unknown displacements at the free DOFs,
# compute the reactions at restrained DOFs, and then
# reconstruct the complete global displacement and force vectors.

rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(
    u_f,
    u_r,
    free_dofs,
    restrained_dofs
)

f_global_complete = assemble_global_forces(
    f_f,
    F_r,
    free_dofs,
    restrained_dofs
)
```

In [60]:

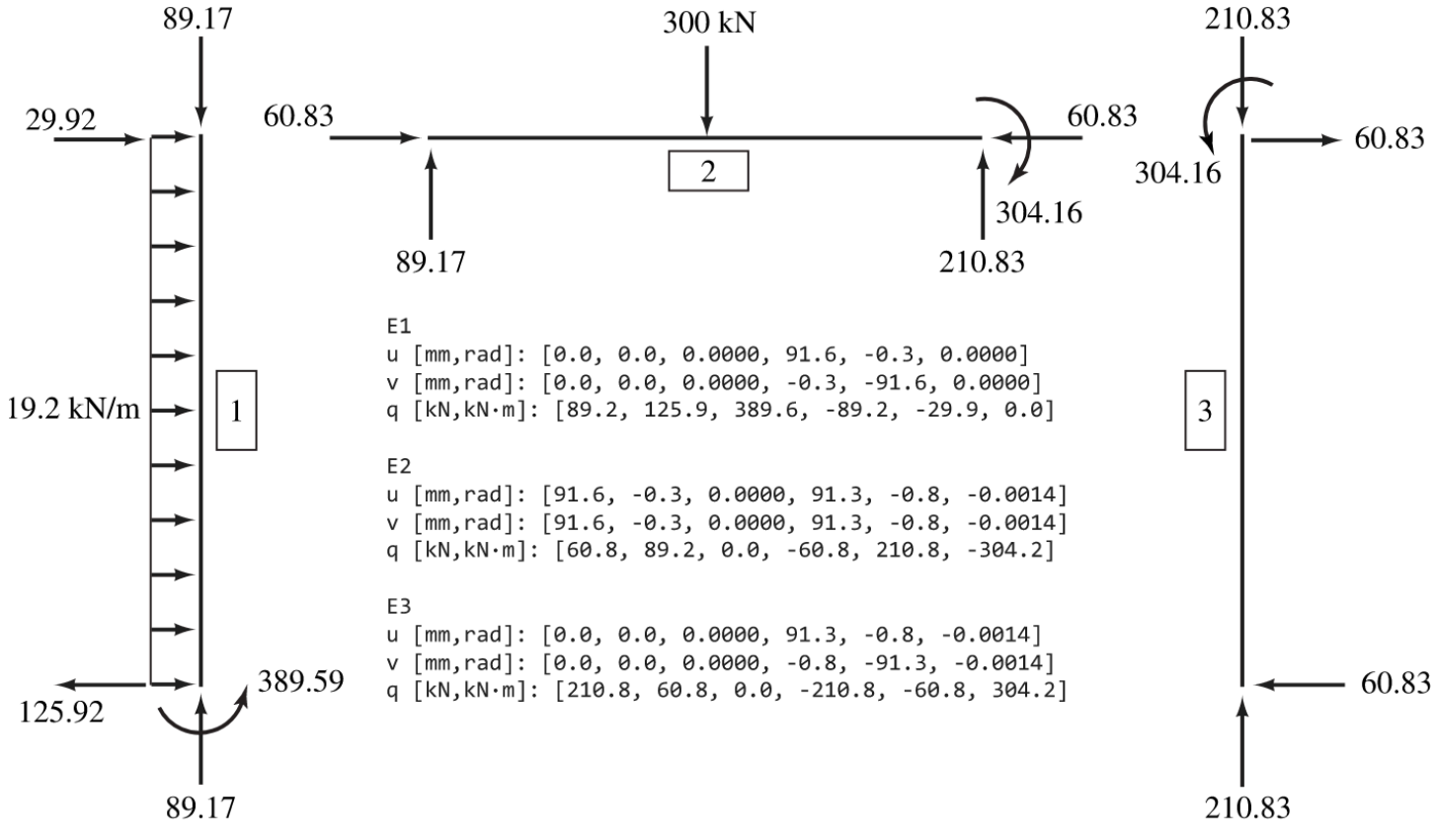
```
# -------------------------
# Display DSM Results
# -------------------------
# Print the final global displacement vector and force vector.

print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
 DOF  Type     Status  Disp (mm / rad)  Load (kN / kN·m)
   1   u_x      Fixed            0.000          -125.918
   2   u_y      Fixed            0.000            89.168
   3 theta      Fixed            0.000           389.592
   4   u_x       Free           91.553            90.750
   5   u_y       Free           -0.343             0.000
   6 theta Fictitious            0.000             0.000
   7   u_x       Free           91.319             0.000
   8   u_y       Free           -0.811             0.000
   9 theta       Free           -0.001             0.000
  10   u_x      Fixed            0.000           -60.832
  11   u_y      Fixed            0.000           210.832
  12 theta Fictitious            0.000             0.000
```

In [61]:
```
# -------------------------
# Display Element Results
# -------------------------
# Compute and print the element-level results for each member.

print_element(1, u_global, m1, T1, k1, Qf1, disp_in_mm=True, dec=1)
print_element(2, u_global, m2, T2, k2, Qf2, disp_in_mm=True, dec=1)
print_element(3, u_global, m3, T3, k3, Qf3, disp_in_mm=True, dec=1)
```

```
E1
u [mm,rad]: [0.0, 0.0, 0.0000, 91.6, -0.3, 0.0000]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.3, -91.6, 0.0000]
q [kN,kN·m]: [89.2, 125.9, 389.6, -89.2, -29.9, 0.0]

E2
u [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
q [kN,kN·m]: [60.8, 89.2, 0.0, -60.8, 210.8, -304.2]

E3
u [mm,rad]: [0.0, 0.0, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.8, -91.3, -0.0014]
q [kN,kN·m]: [210.8, 60.8, 0.0, -210.8, -60.8, 304.2]
```

Solution




```
E1
u [mm,rad]: [0.0, 0.0, 0.0000, 91.6, -0.3, 0.0000]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.3, -91.6, 0.0000]
q [kN,kN·m]: [89.2, 125.9, 389.6, -89.2, -29.9, 0.0]

E2
u [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
q [kN,kN·m]: [60.8, 89.2, 0.0, -60.8, 210.8, -304.2]

E3
u [mm,rad]: [0.0, 0.0, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.8, -91.3, -0.0014]
q [kN,kN·m]: [210.8, 60.8, 0.0, -210.8, -60.8, 304.2]
```

Part 7 - In-Class Excercise

Open `InClass_Excercise.ipynb` in the `../../Code/L8` folder

Part 8 — Implementing Method B

In [62]:

```
import numpy as np
import pandas as pd

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6          # kPa
A = 0.0065         # m^2
I = 150e-6         # m^4
L = 5.0            # m
```

In [63]:

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [64]:

```
# -------------------------
# Member 2 (MT=0)
# -------------------------

factor = (E * I) / (L**3)

k2 = factor * np.array(
    [
        [ (A*L**2)/I,  0.0,      0.0,      -(A*L**2)/I,  0.0,      0.0],
        [ 0.0,         12.0,     6.0*L,    0.0,         -12.0,     6.0*L],
        [ 0.0,         6.0*L,    4.0*L**2, 0.0,         -6.0*L,    2.0*L**2],
        [-(A*L**2)/I,  0.0,      0.0,       (A*L**2)/I,  0.0,      0.0],
        [ 0.0,        -12.0,    -6.0*L,    0.0,          12.0,    -6.0*L],
        [ 0.0,         6.0*L,    2.0*L**2, 0.0,         -6.0*L,    4.0*L**2],
    ],
    dtype=float,
)

# Local fixed-end force vector (MT=0, no releases)
Qf2 = np.array([0.0, 150.0, 187.5, 0.0, 150.0, -187.5], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [65]:

```
# -------------------------
# Member 3 (MT=0)
# -------------------------
k3 = k2.copy()

Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [66]:

```
# -------------------------
# Initialize Global System
# -------------------------
# This slide sets up the global variables needed for assembly of the
# Direct Stiffness Method system.

k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))

# Initialize Applied Load
F_global = np.zeros(ndof, dtype=float)

# Initialize Global Displacement
u_global = np.zeros(ndof, dtype=float)
```

In [67]:

```
# -------------------------
# User-Specified Boundary Conditions
# -------------------------
# This slide defines the external loads and support conditions.

# Applied Load
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)
```

In [68]:
```
# -------------------------
# Assemble Global Stiffness System
# -------------------------
# This step assembles the global stiffness matrix and the global
# fixed-end force vector from all elements.


K_global, F_fef_global = assemble_global_stiffness_and_fef(
    ndof,
    k_list,
    T_list,
    Qf_list,
    map_list
    )
```

In [69]:

```
# -------------------------
# Partition Global System
# -------------------------
# The global stiffness equation is partitioned into free and restrained
# degrees of freedom based on the support conditions.


(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    u_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    u_global,
    F_fef_global,
    dof_restrained_1based
)
```

In [70]:

```
# -------------------------
# Solve Partitioned System
# -------------------------
# Solve for the unknown displacements at the free DOFs,
# compute the reactions at restrained DOFs, and then
# reconstruct the complete global displacement and force vectors.

rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(
    u_f,
    u_r,
    free_dofs,
    restrained_dofs
    )

f_global_complete = assemble_global_forces(
    f_f,
    F_r,
    free_dofs,
    restrained_dofs
    )
```

In [71]:
```
# -------------------------
# Display DSM Results
# -------------------------
# Print the final global displacement vector and force vector.


print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
 DOF  Type Status  Disp (mm / rad)  Load (kN / kN·m)
   1   u_x  Fixed            0.000          -125.918
   2   u_y  Fixed            0.000            89.168
   3 theta  Fixed            0.000           389.592
   4   u_x   Free           91.553            90.750
   5   u_y   Free           -0.343             0.000
   6 theta   Free           -0.007             0.000
   7   u_x   Free           91.319             0.000
   8   u_y   Free           -0.811             0.000
   9 theta   Free           -0.001             0.000
  10   u_x  Fixed            0.000           -60.832
  11   u_y  Fixed            0.000           210.832
  12 theta   Free           -0.027             0.000
```

In [72]:
```
# -------------------------
# Display Element Results
# -------------------------
# Compute and print the element-level results for each member.

print_element(1, u_global, m1, T1, k1, Qf1, disp_in_mm=True, dec=1)
print_element(2, u_global, m2, T2, k2, Qf2, disp_in_mm=True, dec=1)
print_element(3, u_global, m3, T3, k3, Qf3, disp_in_mm=True, dec=1)
```

```
E1
u [mm,rad]: [0.0, 0.0, 0.0000, 91.6, -0.3, -0.0073]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.3, -91.6, -0.0073]
q [kN,kN·m]: [89.2, 125.9, 389.6, -89.2, -29.9, 0.0]

E2
u [mm,rad]: [91.6, -0.3, -0.0073, 91.3, -0.8, -0.0014]
v [mm,rad]: [91.6, -0.3, -0.0073, 91.3, -0.8, -0.0014]
q [kN,kN·m]: [60.8, 89.2, 0.0, -60.8, 210.8, -304.2]

E3
u [mm,rad]: [0.0, 0.0, -0.0267, 91.3, -0.8, -0.0014]
v [mm,rad]: [0.0, 0.0, -0.0267, -0.8, -91.3, -0.0014]
q [kN,kN·m]: [210.8, 60.8, 0.0, -210.8, -60.8, 304.2]
```

Wrap-Up

In this lecture, we:

- Defined **member releases** and enforced hinge conditions ($M_b = 0$ and/or $M_e = 0$)
- Derived modified member relations for:
    - **MT = 1** (hinge at b)
    - **MT = 2** (hinge at e)
    - **MT = 3** (hinges at both ends)
- Discussed **hinged joints** and why they can make $\mathbf{K}$ singular

Next up:

- Secondary effects (support settlement, temperature, fabrication errors)
- How they enter DSM through equivalent nodal effects / modified $\mathbf{Q}_f$